Statement of Purpose

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Research Interests. My primary research interests are interdisciplinary and lie at the intersection of **computer vision (CV) and machine learning (ML), with applications in healthcare and scientific discovery.** My long-term research goal is to leverage the recent advances in **representation learning**, **generative modeling** and **geometric deep learning** (GDL) to develop robust, reasoning capable, and interpretable ML methods and AI systems dedicated to transforming healthcare, particularly in precision medicine, and drug discovery. Today, large web-scale pretrained models, such as, large language models (LLMs) and foundation models (FMs) have emerged as tools to solve problems across domains. However, in scientific domains where the data is scarce, noisy and fragmented, these models quickly hit a wall. Moreover, their "black box" nature and unreliability make them difficult to adapt to high-stakes clinical tasks, where the reasoning behind a prediction is equally, if not, more important.

These challenges motivate my proposed research, which I will investigate through three interconnected themes: (1) **Multi-modal learning**: Developing novel multi-modal architectures that can jointly reason over diverse biological data, *e.g.*, 2D/3D medical images, videos, clinical reports, and genomic profiles, (2) **Geometric Learning**: Developing GDL methods that can represent complex biological shape variations by exploiting the topology of the underlying interactions, (3) **Generative Modeling**: Exploring generative models to simulate "what if" counterfactuals, *e.g.*, predicting how a novel therapeutic intervention will alter a patient's disease trajectory.

Research Experience. My training as a researcher is uniquely interdisciplinary. While my undergraduate was in metallurgical and materials science, I discovered ML in my sophomore year during my involvement in the Vision Language Group at **IIT Roorkee**. I was particularly drawn to computer vision and its applications in biomedical imaging and disease prognosis. It was only after my first taste of research, that I really found my spark.

During the spring of my junior year, I collaborated with Prof. Jose Dolz at ETS Montreal to conduct research in deep learning (DL) based **biomedical image segmentation**. I developed a novel attention mechanism to address the shortcomings of CNNs in medical image segmentation, namely, redundant use of information in multi-scale architectures, and lack of efficient modeling of long-range feature dependencies. Specifically, we proposed a multi-scale, stacked, self-guided attention module to effectively integrate local features with their corresponding global dependencies, resulting in modeling of features at multiple scales, and progressive weighing of importance semantic regions while masking out irrelevant noise. Our work is published in the **Journal of Biomedical and Health Informatics** [1], and has garnered **over 630 citations**. This project was pivotal, teaching me how to take a project from an idea to an accepted publication, quickly reproduce baselines, produce clean open-source code, and communicate and collaborate effectively.

I further explored **generative modeling for biomedical applications** during my junior year summer internship at Preferred Networks (PFN), where I was one of the 10 students selected as part of PFN global internship program. My project aimed to reduce patients' radiation exposure by developing a vector-quantization-based GAN to reconstruct 3D CT volumes from orthogonal 2D X-rays. Over the course of the summer, I learned the valuable skills of running and managing large-scale experiments, adaptability since I had to migrate from pytorch to an unfamiliar internal ML framework, and the impact of ML methods in real-world clinical applications. Our work was presented at a **NeurIPS 2019** workshop [2]. My interactions with fellow interns at PFN and senior researchers at NeurIPS broadened my outlook on research and motivated me to pursue graduate studies.

To gain more research experience and explore **geometric deep learning**, I worked with Prof. Jonghyun Choi on unsupervised **multi-target domain adaptation (MTDA) of 3D point clouds**. While single-target domain adaptation (STDA) was well studied in the literature, MTDA was barely explored for 3D vision tasks, despite its wide real-world applications in autonomous driving. To address this shortcoming, I proposed MEnSA [3], a simple yet effective ensemble-average-based mixup approach that uses implicit adversarial training to adapt a single source model to multiple target domains. This work, initially submitted to NeurIPS 2021 but finally accepted at a **CVPR 2023** workshop taught me an important lesson: the

tenacity as a researcher to continually incorporate feedback and improve the work, a quality I now see as essential for a successful PhD.

Building on this experience, I joined Prof. Ghassan Hamarneh's lab at **Simon Fraser University** (SFU) for my MSc in Computer Science, and explored **3D computer vision for biomedical applications**. My first thesis project, DermSynth3D [4], tackled the critical **data scarcity in dermatology**. We developed a data generation framework to create photorealistic *in-the-wild* synthetic dermatological data with rich multi-task annotations for various skin-analysis tasks. Specifically, our framework blends skin disease patterns onto 3D textured meshes of human subjects using a differentiable renderer and generates 2D images from various camera viewpoints under chosen lighting conditions in diverse background scenes. As a co-first author, I particularly contributed in the design and implementation of the core framework, data curation, and designing and conducting experiments. This work is published in **Medical Image Analysis** journal [4].

My second project, TrIND [5], addressed the challenge of accurately representing the topologically complex **anatomical structures**. In this work, I proposed a novel approach for representing anatomical trees using **implicit neural representations** (INRs). Further, I developed a **denoising diffusion** model to capture the distribution of a set of trees in the INR latent space, allowing for high-fidelity generation. This work got accepted in **MICCAI** 2024.

Upon graduation, I joined Huawei's Noah's Ark Lab to conduct research in **vision language models (VLMs)** for embodied AI. I worked with Dr. Binbin Xu on developing real-time zero-shot pose estimation methods, optimizing models for latency and memory. I also mentored an intern on a project, now accepted in **CoRL 2025** [6], where we exploited priors and uncertainty estimates from pretrained diffusion models for **zero-shot pose estimation**. This industry experience was invaluable, but it also crystalized my core motivation: I realized my true passion lies not just in solving complex ML problems, but in applying them to foundational challenges in healthcare and scientific discovery. Towards this goal, I recently joined Alberta Machine Intelligence Institute (Amii) to develop efficient and accurate models for **cell detection** in high-res microscopy images. My experience and innovative thinking developed while conducting research in both academic and industry settings, complemented with rich insights derived from a range of projects, has further deepened my passion for research. Therefore, I believe a PhD is the natural step for me to grow as an independent researcher.

Why UCB? Mention a couple of professors that you want to work with, and how you're research aligns with theirs, citing some concrete examples (like their recent works), and why you think you're a good fit to be working with them. Don't just name drop.

References

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